

Endoscopic Plantar Partial Sesamoidectomy for the Management of Metatarsalgia of the First Metatarsal Head After Arthrodesis of the First Metatarsophalangeal Joint

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Abstract: First metatarsophalangeal arthrodesis has been used to treat various pathologies of the great toe with good clinical outcome. First metatarsal head metatarsalgia can occur after first metatarsophalangeal arthrodesis. Surgery is indicated if conservative measures cannot alleviate the symptoms. If there is isolated excessive plantar pressure of the first metatarsophalangeal joint without significant sagittal deformity of the first metatarsophalangeal joint, plantar partial ostectomy of the sesamoid bone(s) is indicated. The purpose of this technical note is to describe the technical details of endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint. This all-endoscopy approach allows partial ostectomy of the sesamoid bone from its plantar side dorsally and is a more direct approach than arthroscopic sesamoidectomy to eliminate the plantar pressure point. The extent of partial ostectomy can be titrated according to the intraoperative assessment of plantar pressure distribution across the metatarsal heads. The aim is to eliminate the pressure point underneath the first metatarsal head without excessive bone removal and create excessive pressure underneath the second metatarsal head.

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First metatarsophalangeal arthrodesis has been used to treat various pathologies of the great toe with good clinical outcome.¹ Excessive plantar pressure can occur under the first metatarsal head after the operation, and it is frequently because of excessive plantarflexion of the first metatarsal.¹ Plantar keratosis may develop

when the sesamoid bone excessively protrudes toward the plantar side.^{2,3} This first metatarsal head metatarsalgia after first metatarsophalangeal arthrodesis can be treated conservatively with rest, ice, orthotics, activity modification, and nonsteroidal anti-inflammatory medications.⁴ Surgery is indicated if conservative measures

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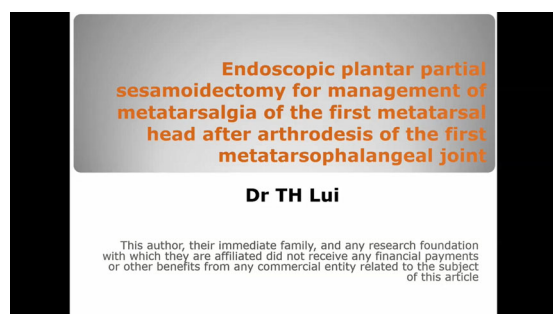
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VIDEO 1 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. This is performed via the proximal and distal portal which are interchangeable as the viewing and working portals. The flexor hallucis longus tendon is the anatomic landmark of this endoscopic approach. The soft tissue covering the medial sesamoid is resected with an arthroscopic shaver, and then, partial ostectomy is performed from the edge toward the body of the sesamoid bone. After completion of partial sesamoidectomy, the integrity of the flexor hallucis longus tendon is confirmed endoscopically. Video content can be viewed at <https://doi.org/10.1002/atn2.70308>.

cannot alleviate the symptoms. Corrective osteotomy is indicated if the first metatarsophalangeal joint is fused in excessive dorsiflexion position, resulting in excessive plantarflexion of the first metatarsal.¹ If there is isolated excessive plantar pressure of the first metatarsophalangeal joint without significant sagittal deformity of the first metatarsophalangeal joint, plantar partial ostectomy of the sesamoid bone(s) is indicated.³ Open sesamoidectomy is effective to relieve the plantar pressure but is associated with an overall complication rate of 22.5%.² In general, the complications of sesamoidectomy

include hallux varus or valgus deformity, plantarflexion weakness and claw hallux deformity, painful plantar incision, injury to the ligaments and cartilage of the first metatarsophalangeal joint, decreased range of motion of the first metatarsophalangeal joint, transfer metatarsalgia, and digital nerve injury.^{2,4} To minimize the surgical risks, techniques of arthroscopic sesamoidectomy have been developed.^{1,5,6} A fluoroscopic- and endoscopic-assisted plantar partial ostectomy of medial sesamoid has also been reported, in which plantar partial ostectomy is performed under fluoroscopic guidance and, after that, endoscopy is performed to confirm the integrity of the flexor hallucis longus (FHL) tendon.³ We have developed an all-endoscopy technique of partial ostectomy of the plantar part of the medial sesamoid. The purpose of this technical note is to describe the technical details of endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint. This endoscopic approach allows partial ostectomy of the sesamoid bone from its plantar side dorsally and is a more direct approach than arthroscopic sesamoidectomy to eliminate the plantar pressure point. It is because endoscopic approach directly removes bone from the plantar side of the sesamoid bone. In contrast, the arthroscopic approach removes bone from the dorsal side of the sesamoid bone and then dorsal shift of the sesamoid bone. It is indicated for first metatarsal head metatarsalgia because of excessive plantar pressure at the fused first metatarsophalangeal joint with relatively normal sagittal alignment of the great toe.¹ It is contraindicated if there is other source of pain, for example, metatarsosesamoid synovitis or degeneration, first metatarsal head metatarsalgia with excessive metatarsal plantarflexion, or previous excision of a sesamoid or absence of a sesamoid on the same foot (Table 1).⁷

TABLE 1 Indications and Contraindications of Endoscopic Plantar Partial Sesamoidectomy for the Management of Metatarsalgia of the First Metatarsal Head After Arthrodesis of the First Metatarsophalangeal Joint

Indications	Contraindications
1. First metatarsal head metatarsalgia because of excessive plantar pressure at the fused first metatarsophalangeal joint with relatively normal sagittal alignment of the great toe	1. There is other source of pain, for example, metatarsosesamoid synovitis or degeneration
	2. First metatarsal head metatarsalgia with excessive metatarsal plantarflexion
	3. Previous excision of a sesamoid or absence of a sesamoid on the same foot

SURGICAL TECHNIQUE

Preoperative Planning and Patient Positioning

Preoperative clinical determination of the painful site is important and should be confirmed by local tenderness at the medial sesamoid bone and static and dynamic pressure studies. Standing lateral radiograph of the foot is also helpful for assessing the degree of plantarflexion of the first metatarsal and the degree of plantar displacement of the medial sesamoid bone (Figure 1). Preoperative computed tomogram is used for studying the morphology of the medial sesamoid bone, and postoperative computed tomogram can assess the amount of bone resection (Figure 2).

The patient is placed in the supine position with the legs spread. A 2.7-mm 30° arthroscope (Henke Sass Wolf GmbH, Tuttlingen, Germany) is used for this procedure. A thigh tourniquet is applied to provide a bloodless operative field. Fluid inflow is driven by gravity, and no arthropump is used.

Portal Placement

The procedure is performed via the proximal and distal portals. The proximal portal is about 2 cm proximal-medial to the medial sesamoid. The distal portal is about 2 cm distal to the medial sesamoid and over the FHL tendon (Figure 3). Four-millimeter skin incisions are made at portal sites. The subcutaneous tissue is bluntly dissected with a hemostat, and endoscopic working space is created

at the plantar surfaces of the medial sesamoid and FHL tendon. This endoscopic working space orientates from proximal-dorsal-medial to distal-plantar-lateral and connects the proximal and distal portals (Figure 4). The FHL tendon is the anatomic landmark of this endoscopic procedure.

Plantar Partial Sesamoidectomy of Lateral Half of the Medial Sesamoid

The proximal portal is the viewing portal, and the distal portal is the working portal. The FHL is identified, and the medial sesamoid is just medial to it. The soft tissue covering the lateral half of the medial sesamoid is resected with an arthroscopic shaver (Dyonics, Smith & Nephew, Andover, MA). After that, the exposed sesamoid bone is shaved with an arthroscopic burr (Dyonics) from the lateral edge toward the body of the sesamoid bone (Figure 5). During the procedure, frequent assessment of the plantar pressure of the resection area is performed to ensure that it is similar to the plantar pressure underneath the second metatarsal head.

Plantar Partial Sesamoidectomy of Medial Half of the Medial Sesamoid

The distal portal is the viewing portal, and the proximal portal is the working portal. The FHL is identified, and the medial sesamoid is just medial to it. The soft tissue covering the medial half of the medial sesamoid is resected with the arthroscopic shaver. After that, the exposed sesamoid



FIGURE 1 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. Standing radiograph of the illustrated case's left foot showed that the MS abuts on the ground. (MS, medial sesamoid.)

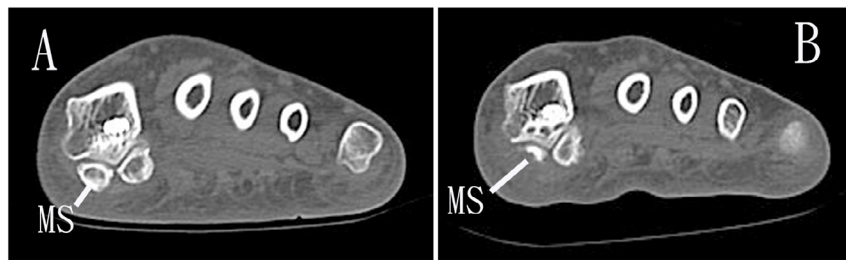


FIGURE 2 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. (A) Preoperative coronal CT image shows that the MS is more plantarly placed than the lateral sesamoid, and it has a relatively sharp plantar angle. (B) Postoperative coronal CT image shows that the plantar part of the MS is resected. (CT, computed tomography; MS, medial sesamoid.)

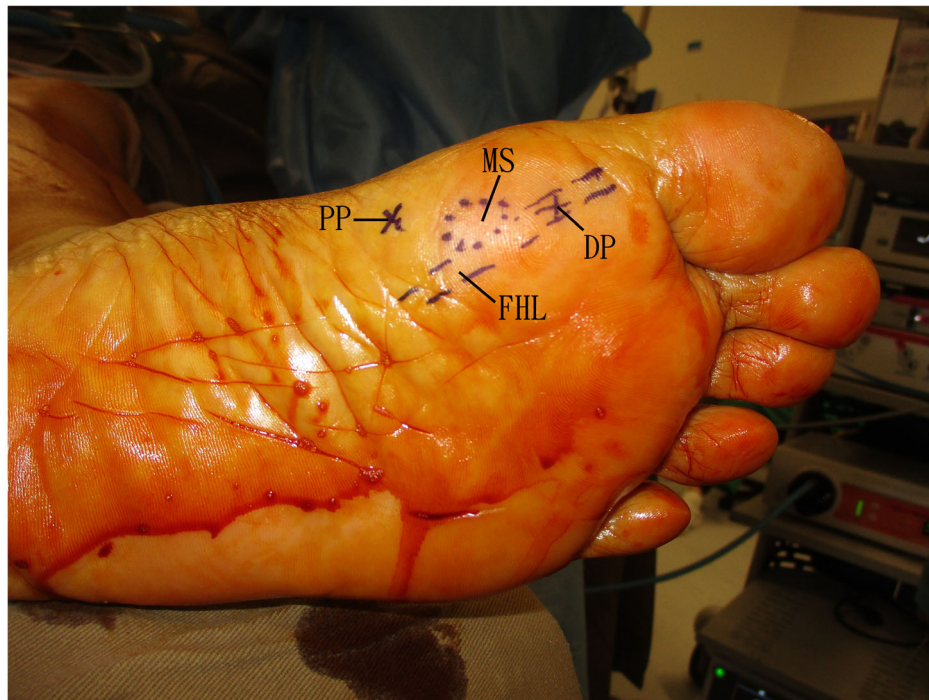


FIGURE 3 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. The procedure is performed via the proximal and distal portals. The proximal portal is about 2 cm proximal-medial to the medial sesamoid. The distal portal is about 2 cm distal to the medial sesamoid and over the flexor hallucis longus tendon. (DP, distal portal; FHL, flexor hallucis longus; MS, medial sesamoid; PP, proximal portal.)

bone is shaved with the arthroscopic burr from the medial edge toward the body of the sesamoid bone (Figure 6).

Assessment of the Amount of Bone Resection and Integrity of the FHL Tendon After the Procedure

The proximal portal is the viewing portal. The amount of bone resection and integrity of the FHL is assessed after completion of plantar partial sesamoidectomy (Figure 7, Video 1).

Postoperatively, weight-bearing walking as pain tolerated is allowed. Physiotherapy is prescribed for swelling control and active and progressive mobilization of the first metatarsophalangeal joint.

DISCUSSION

Endoscopic plantar partial sesamoidectomy is an effective minimally invasive surgical approach to manage metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint. Complications of

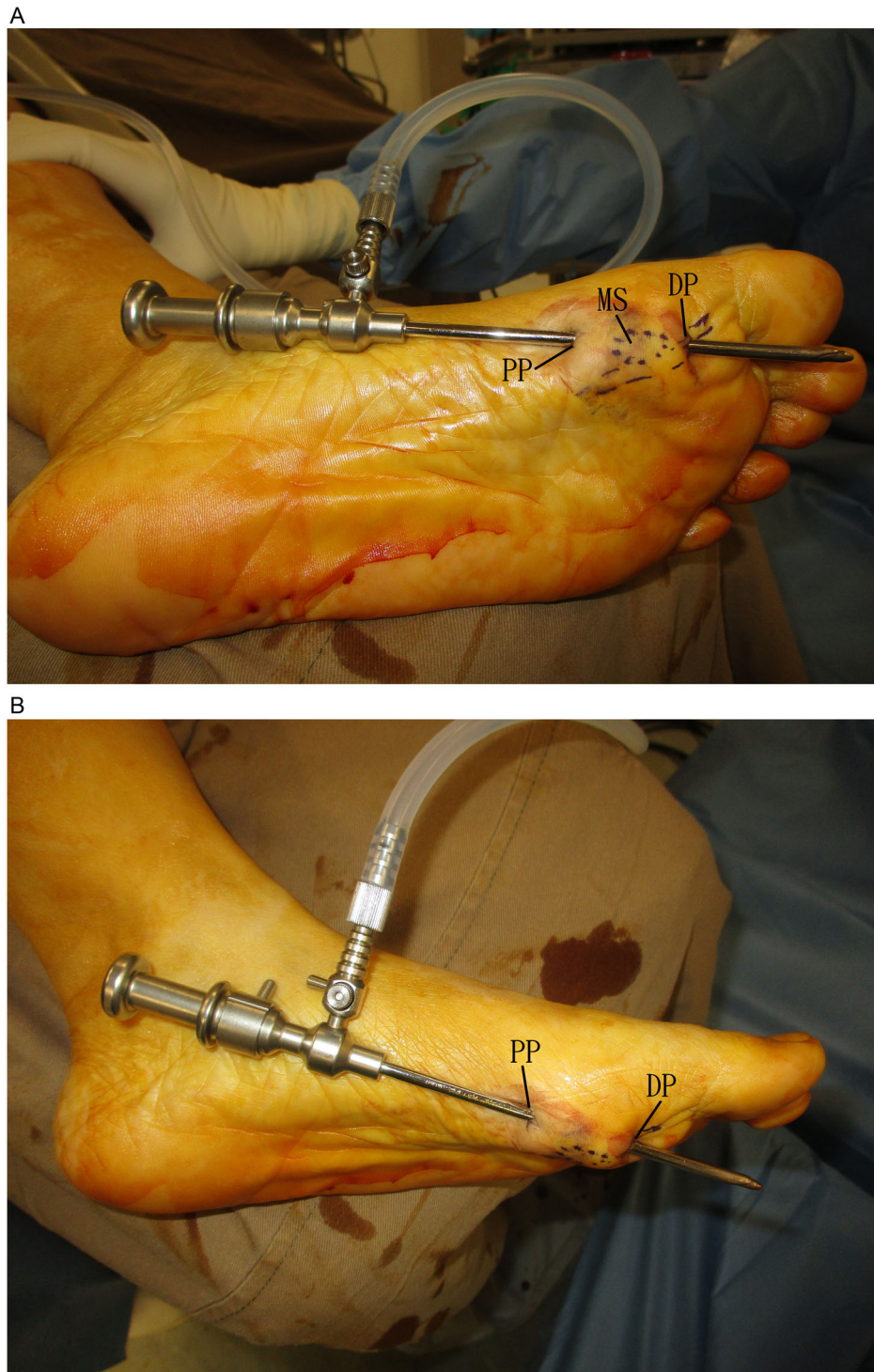


FIGURE 4 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. Endoscopic working space is created at the plantar surfaces of the medial sesamoid and flexor hallucis longus tendon. This endoscopic working space orientates from proximal-dorsal-medial to distal-plantar-lateral and connects the proximal and distal portals: (A) plantar-dorsal view and (B) lateral view. (DP, distal portal; MS, medial sesamoid; PP, proximal portal.)

sesamoidectomy related to development of deformity of the first metatarsophalangeal joint will not occur if sesamoidectomy is performed in patient with fused metatarsophalangeal joint. However, other soft-tissue complications, for

example, digital nerve injury, wound infection or dehiscence, and painful plantar scar formation, can be quite disturbing for the patient. Arthroscopic and endoscopic approaches of sesamoidectomy can reduce these risks.

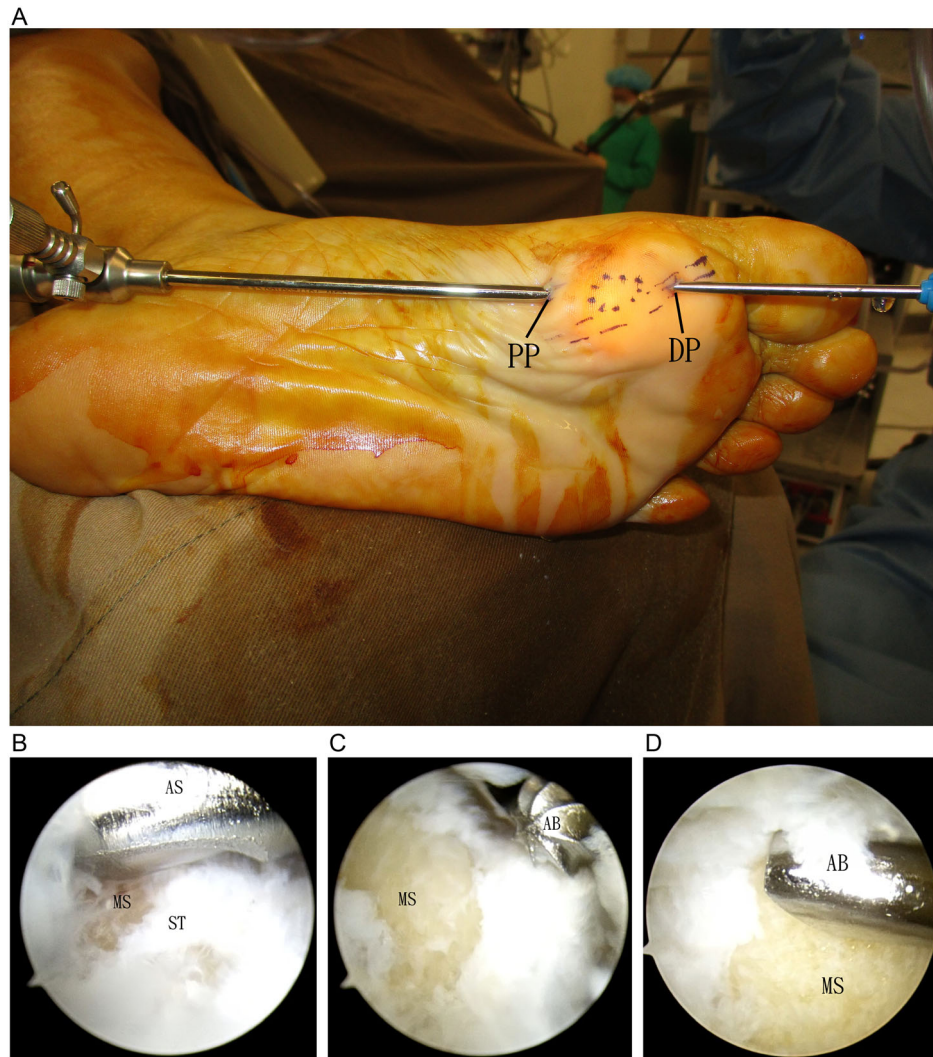


FIGURE 5 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. (A) The proximal portal is the viewing portal, and the distal portal is the working portal. (B) The soft tissue covering the lateral half of the medial sesamoid is resected with an arthroscopic shaver. (C) Partial osteotomy of the lateral edge of the medial sesamoid is performed with an arthroscopic burr. (D) Partial osteotomy of the body of the medial sesamoid is performed with the arthroscopic burr. (AB, arthroscopic burr; AS, arthroscopic shaver; DP, distal portal; MS, medial sesamoid; PP, proximal portal; ST, soft tissue.)

Arthroscopic approach is more familiar approach for orthopaedic surgeons. However, it is a difficult approach because the metatarsosesamoid compartment is usually contracted after previous fusion surgery and the fused metatarsophalangeal joint cannot be plantarflexed to increase the arthroscopic working space. Moreover, the partial osteotomy of the medial sesamoid from dorsal side plantarly may not be effective to relieve the plantar pressure point unless complete sesamoidectomy is performed. However, complete excision in the medial sesamoid is usually unnecessary and may result in transfer metatarsalgia to the lesser metatarsals or “transfer metatarsalgia” to the adjacent fibular sesamoid.

Endoscopic sesamoidectomy works in a virgin surgical field, and the adjacent flexible soft tissue allows more

freedom of instrument manipulation. Moreover, partial osteotomy of the medial sesamoid from plantar side dorsally allows more direct elimination of the plantar pressure point, and the extent of partial osteotomy can be titrated according to the intraoperative assessment of plantar pressure distribution across the metatarsal heads. However, pathology of the metatarsosesamoid articulation cannot be dealt with unless complete sesamoidectomy is performed. The key to success is correct selection of surgical approach by accurate preoperative determination of the source of plantar pain after first metatarsophalangeal fusion.

The goal of this endoscopic procedure is adequate but not excessive plantar partial osteotomy of the medial sesamoid. The extent of partial osteotomy should be

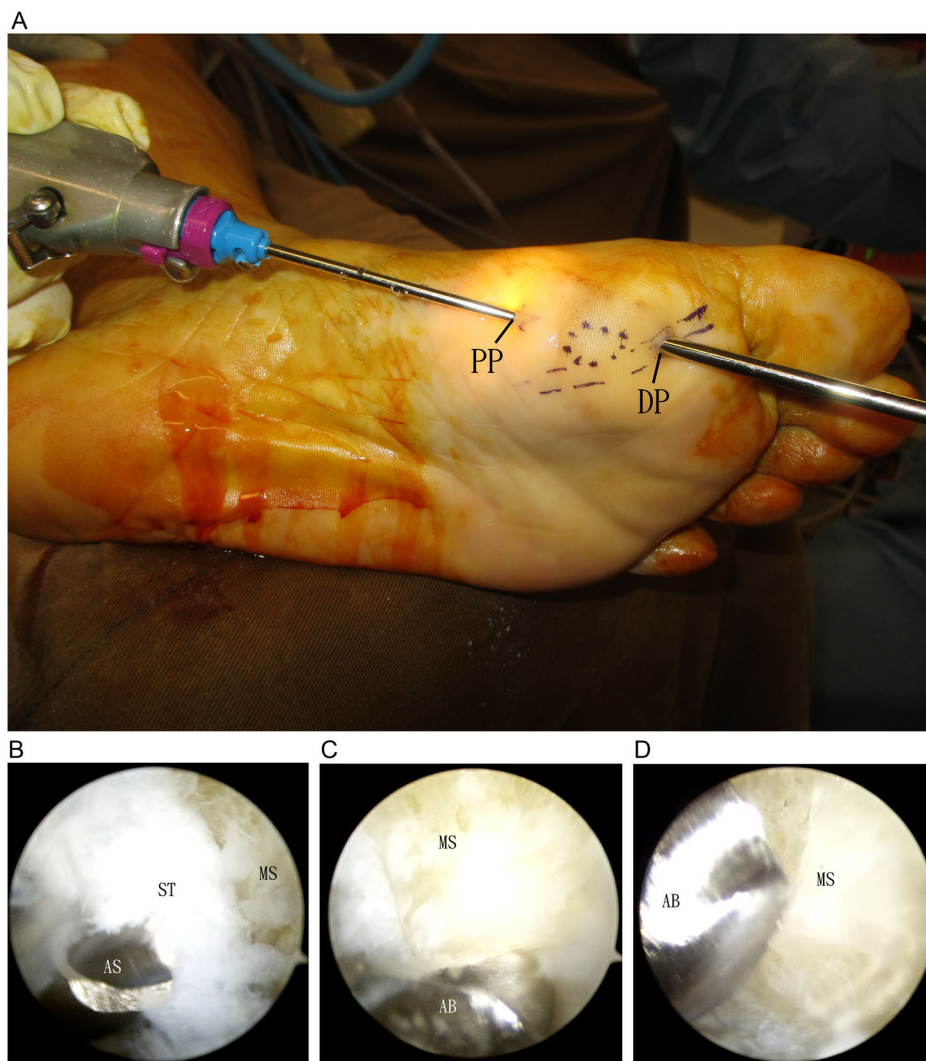


FIGURE 6 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. (A) The distal portal is the viewing portal, and the proximal portal is the working portal. (B) The soft tissue covering the medial half of the medial sesamoid is resected with the arthroscopic shaver. (C) Partial osteotomy of the medial proximal edge of the medial sesamoid is performed with an arthroscopic burr. (D) Partial osteotomy of the body of the medial sesamoid is performed with the arthroscopic burr. (AB, arthroscopic burr; AS, arthroscopic shaver; DP, distal portal; MS, medial sesamoid; PP, proximal portal; ST, soft tissue.)

titrated intraoperatively according to the plantar pressure distribution amount the metatarsal heads. Under-resection of bone may result in persistent pressure point and metatarsalgia underneath the first metatarsal head. It is also important to assess the pressure distribution between the medial and lateral sesamoids by carefully palpating and pressing the sesamoids, to avoid “transfer metatarsalgia” to the lateral sesamoid. To avoid iatrogenic creation of pressure points within the medial sesamoid, any bone spike at the bone cutting surface should be removed and the edges of the medial sesamoid should be adequately shaved. The distal edge of the bone cutting surface should be smoothened to avoid pressure point during terminal stance of walking (Table 2).

The advantages of this minimally invasive approach include better cosmetic results, minimal soft-tissue trauma, decreased postoperative pain, less chance of painful scar formation, and titration of the extent of partial osteotomy according to the intraoperative assessment of pressure distribution across the metatarsal heads.¹ The potential risks include injury to the digital nerve, sesamoid fracture, residual first metatarsal head metatarsalgia, transfer metatarsalgia to second metatarsal, FHL tendon dislocation, rupture or fibrous adhesions, and resultant deformity of the interphalangeal joint of the great toe (Table 3).¹ This approach is not technically difficult and can be managed by the average foot and ankle arthroscopist.

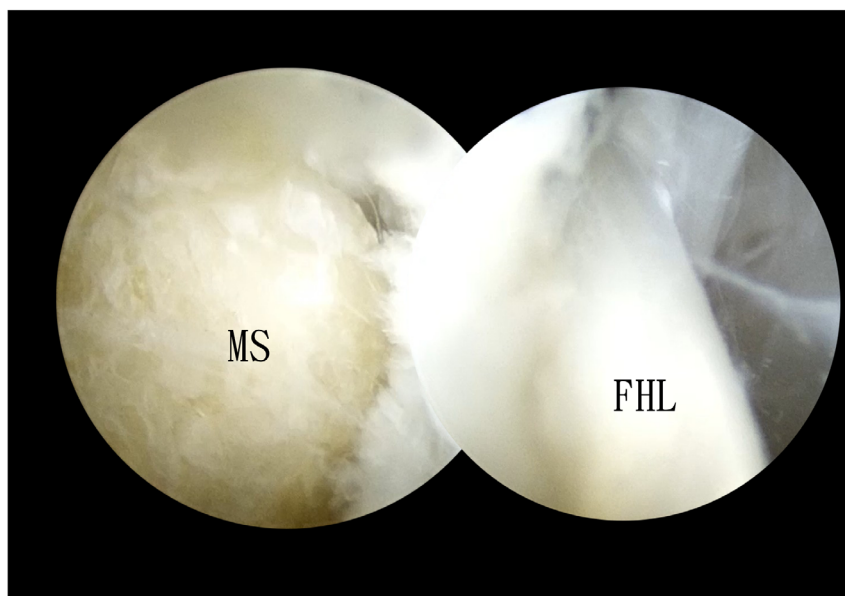


FIGURE 7 Endoscopic plantar partial sesamoidectomy for the management of metatarsalgia of the first metatarsal head after arthrodesis of the first metatarsophalangeal joint of the left foot. The patient is placed in the supine position with the legs spread. The proximal portal is the viewing portal. The amount of bone resection and integrity of the FHL is assessed after completion of plantar partial sesamoidectomy. (FHL, flexor hallucis longus; MS, medial sesamoid.)

TABLE 2 Pearls and Pitfalls of Endoscopic Plantar Partial Sesamoidectomy for the Management of Metatarsalgia of the First Metatarsal Head After Arthrodesis of the First Metatarsophalangeal Joint

Pearls	Pitfalls
1. The flexor hallucis longus tendon is the anatomic landmark of this endoscopic procedure	1. Incorrect preoperative determination of source of pain will lead to persistent pain after this procedure
2. Intraoperative monitoring of the plantar pressure distribution of the metatarsal heads can help to titrate the degree of partial osteotomy needed	2. Excessive partial osteotomy may lead to transfer metatarsalgia
3. Intraoperatively, the plantar pressure distribution between the medial and lateral sesamoids is carefully monitored to avoid excessive partial osteotomy leading to “transfer metatarsalgia” to the lateral sesamoid	
4. Any bone spike at the bone cutting surface and the edges of the medial sesamoid should be adequately shaved	
5. The distal edge of the bone cutting surface should be smoothened	

TABLE 3 Advantages and Risks of Endoscopic Plantar Partial Sesamoidectomy for the Management of Metatarsalgia of the First Metatarsal Head After Arthrodesis of the First Metatarsophalangeal Joint

Advantages	Risks
1. Better cosmetic results	1. Injury to the digital nerve
2. Minimal soft-tissue trauma	2. Sesamoid fracture
3. Decreased postoperative pain	3. Residual first metatarsal head metatarsalgia
4. Less chance of painful scar formation	4. Transfer metatarsalgia to second metatarsal
5. Titration of the extent of partial osteotomy according to the intraoperative assessment of pressure distribution across the metatarsal heads	5. Flexor hallucis longus tendon dislocation, rupture or fibrous adhesions, and resultant deformity of the interphalangeal joint of the great toe

DISCLOSURES

The authors (X.P., T.H.L., C.C.H.L., A.M.Y.S., J.H., X.W., Y.P.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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